Materials Chemistry for Devices and E-Waste Management

CHY1004

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4 CREDITS (IPCC)

Unit I 09Hrs

- Electrodes and Energy Systems

Unit II 09Hrs

- Corrosion Science and Metal Finishing

Unit III 09Hrs

- Analytical Techniques, Memory Devices and Display Systems

Unit IV 09 Hrs

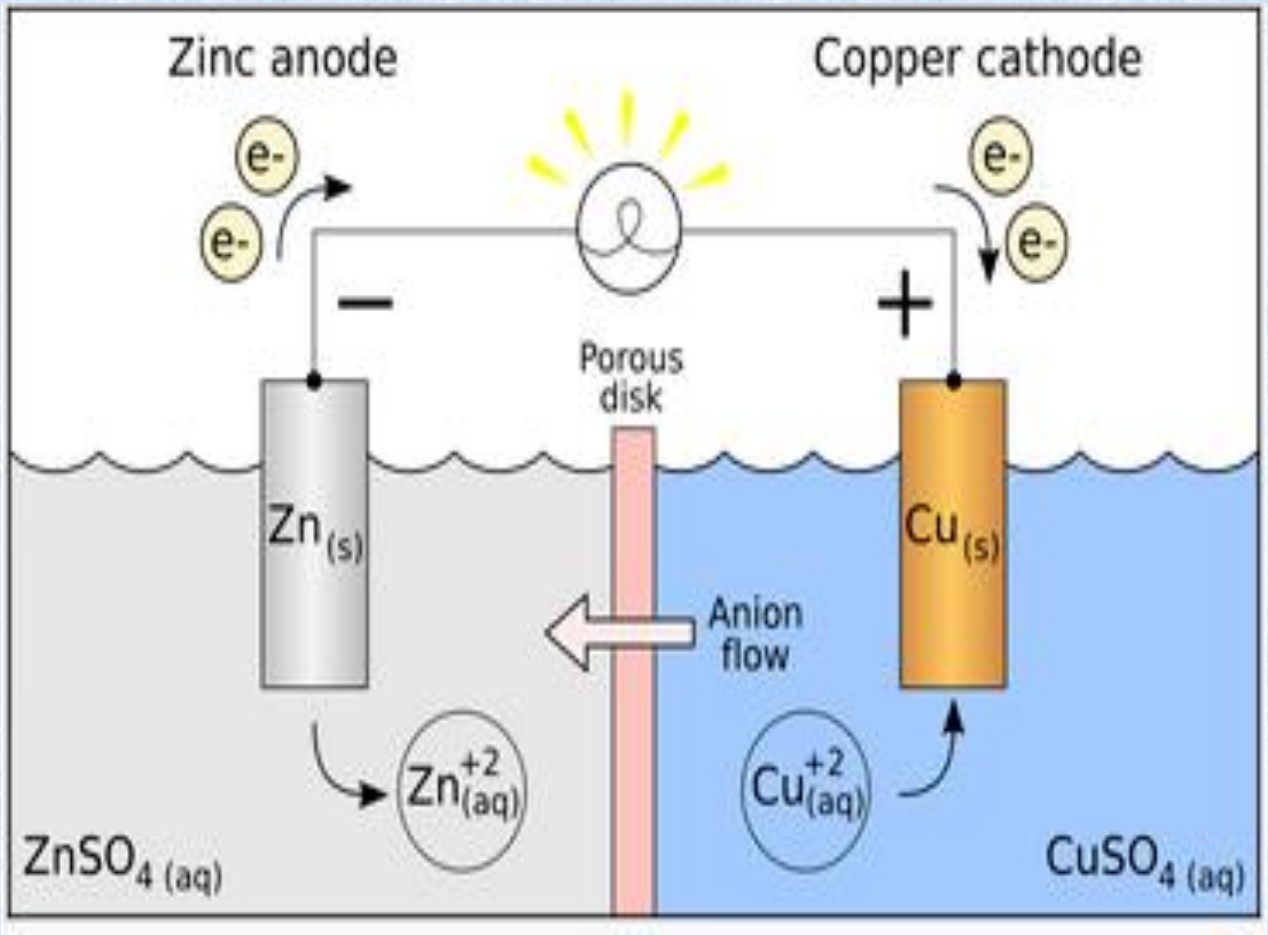
- Polymers and Nanomaterials

Unit V 09 Hrs

- E-Waste Management

Unit-I

ELECTRODE SYSTEM



What do you mean by Electrochemical Cell?

Ions
(A^+ or B^-)

Chemical Energy

Electrochemical Cells are devices which convert Chemical Energy to Electrical Energy or vice versa

Electrons
(e^-)

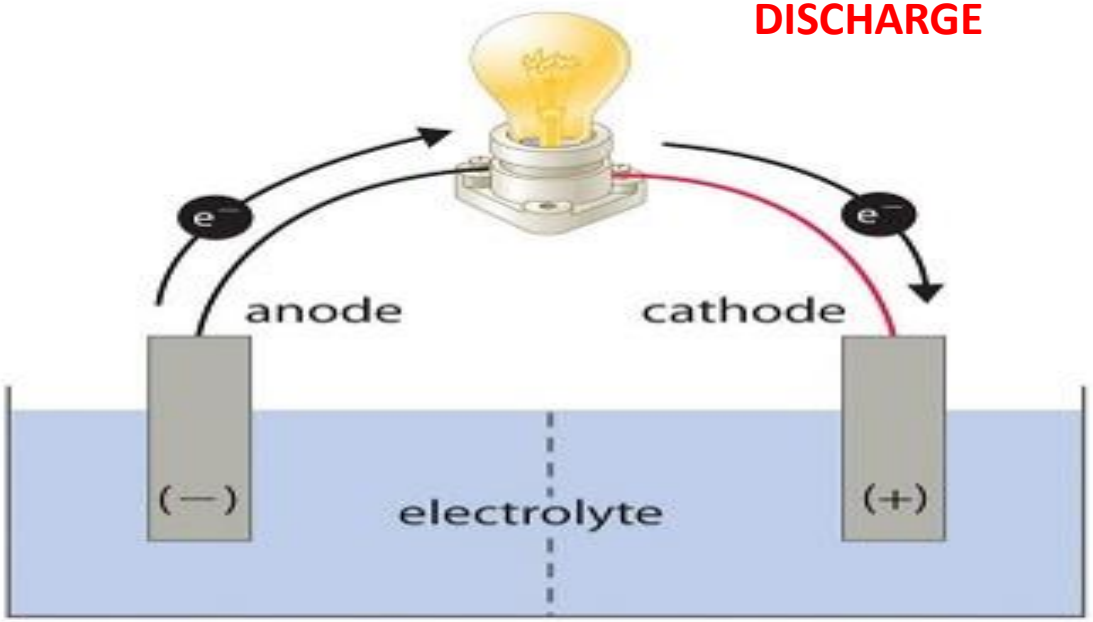
Electrical Energy

Energy is produced by movement of charged species.

Chemical Energy → Electrical Energy	Electrical Energy → Chemical Energy

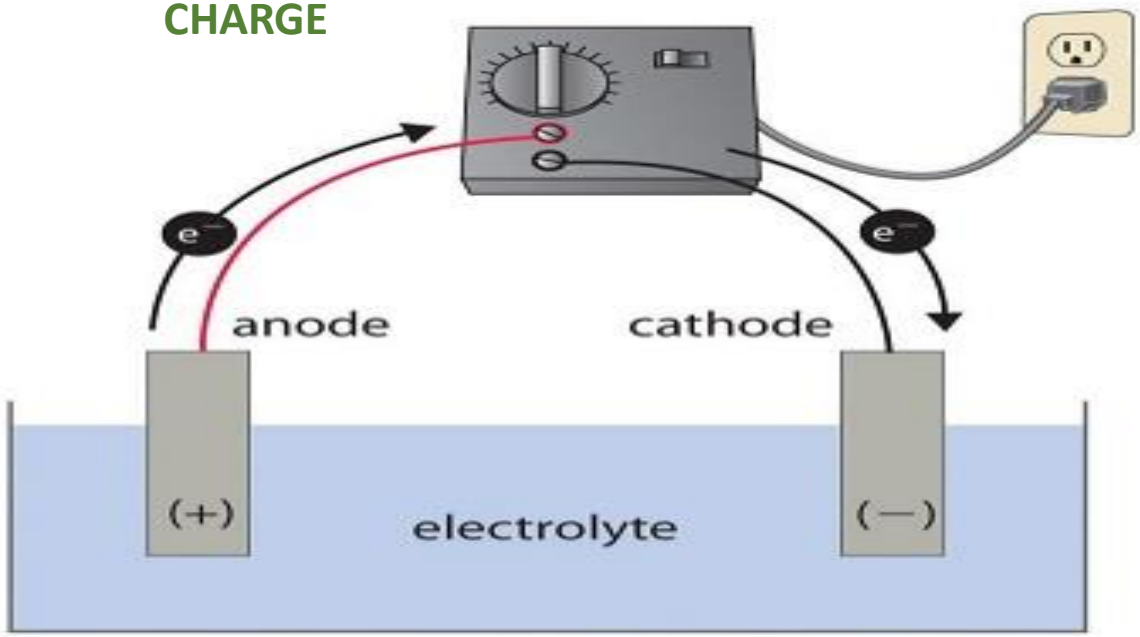
DISCHARGE

CHARGE



GALVANIC CELL

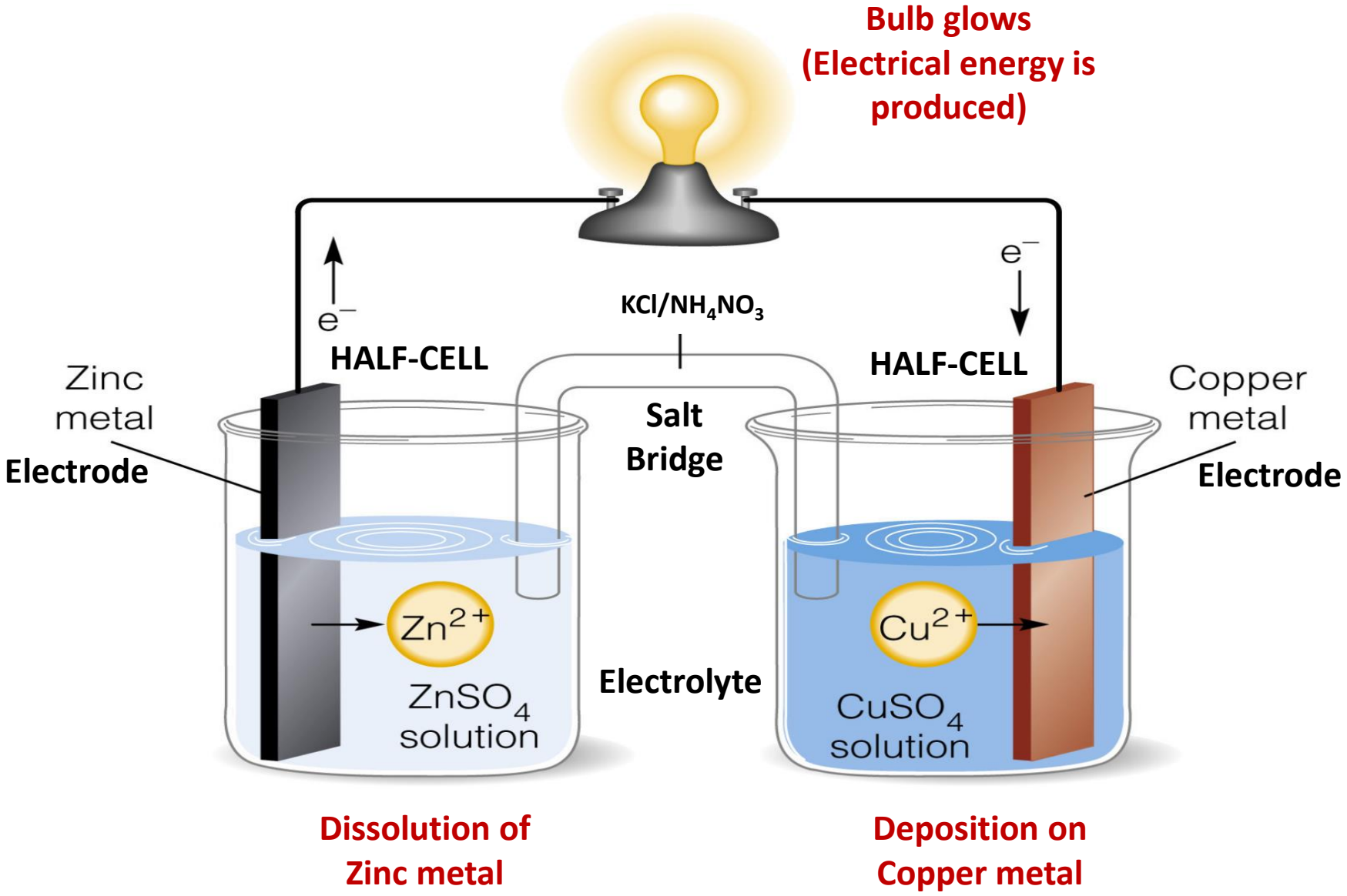
Energy released by spontaneous redox reaction is converted to electrical energy.



ELECTROLYTIC CELL

Electrical energy is used to drive nonspontaneous redox reaction.

DANIEL CELL

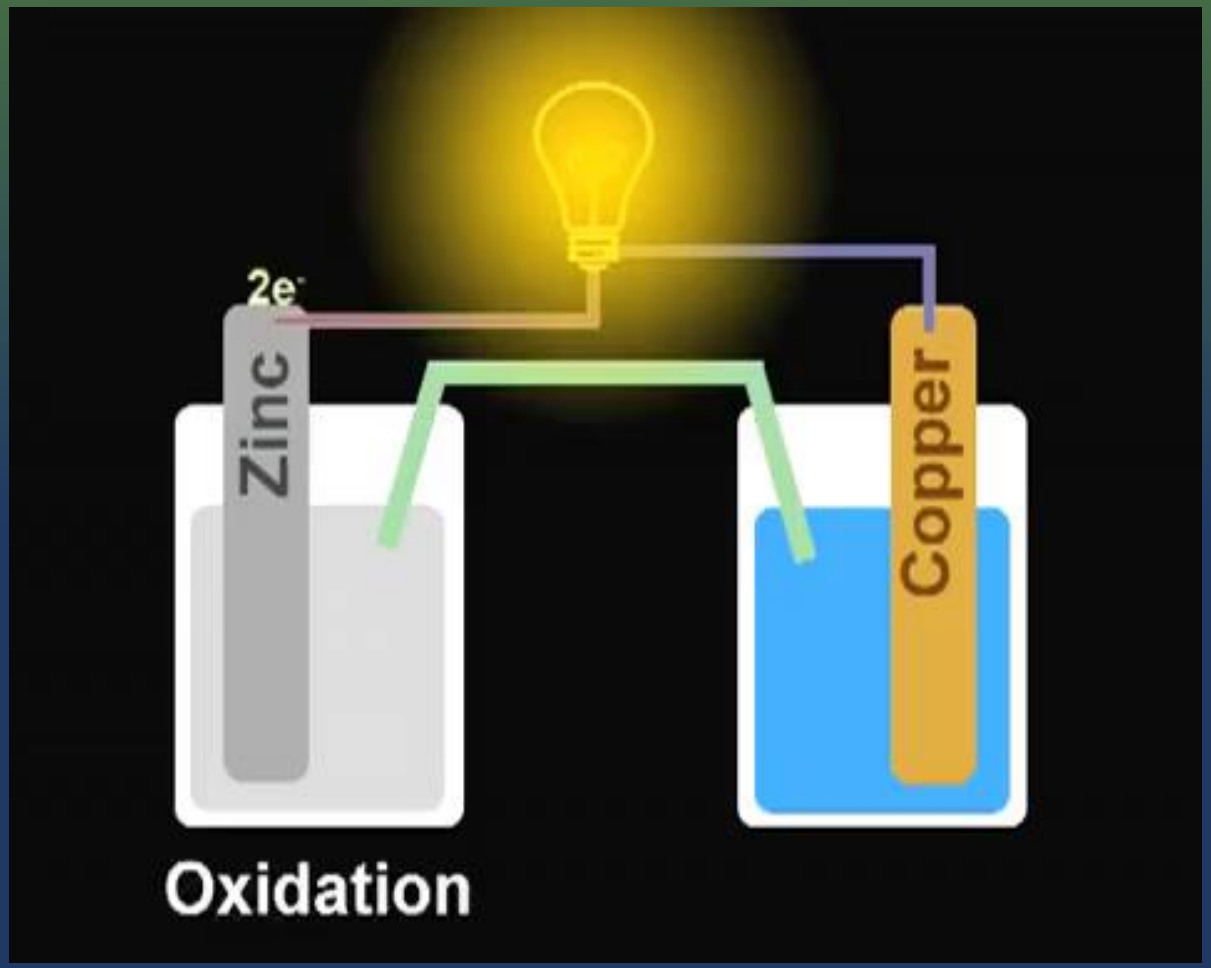

Bulb glows
(Electrical energy is produced)
e⁻
e⁻
KCl/NH4NO3
HALF-CELL
HALF-CELL
Zinc metal
Electrode
Salt Bridge
Copper metal
Electrode
Zn2+
Cu2+
ZnSO4 solution
Electrolyte
CuSO4 solution
Dissolution of Zinc metal
Deposition on Copper metal

$$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$$

$$Zn \rightarrow Zn^{2+} + 2e^-$$

Anode

Loss of e⁻ is oxidation



$$Cu^{2+} + 2e^- \rightarrow Cu$$

Cathode

Gain of e⁻ is reduction

Electrochemical REDOX reaction

At anode: $Zn(s) \rightarrow Zn^{2+}(aq) + \cancel{2e^-}$

At cathode: $Cu^{2+}(aq) + \cancel{2e^-} \rightarrow Cu(s)$

Net cell reaction: $Zn(s) + Cu^{2+}(aq) \rightarrow Cu(s) + Zn^{2+}(aq)$

NO ELECTRONS IN THE NET CELL REACTION

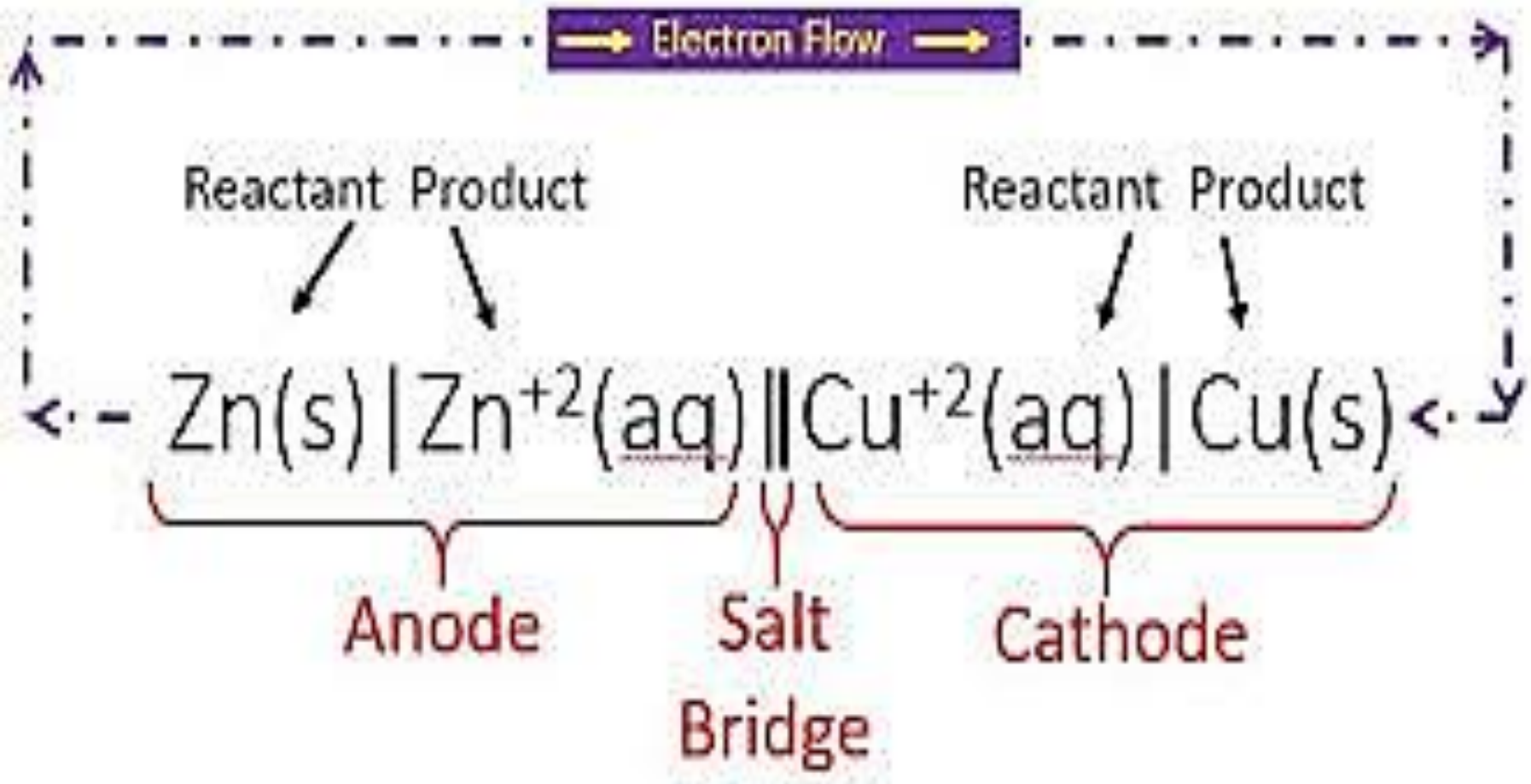
RULES FOR WRITING CELL NOTATION

1) Anode (oxidation) is written first, followed by cathode (reduction).
2) The salt bridge, separating the half-cells, is represented as double vertical line (||).
3) The electrode and electrolyte are separated by a single vertical line (|).
4) In a half-cell, the reactants are written first followed by product.
5) Any information on the strength and/or state of the components is written in a small bracket adjacent to that component.

$$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$$

reactants products

$$Zn \quad / \quad Zn^{2+} \quad // \quad Cu^{2+} \quad / \quad Cu$$


Electron Flow
Reactant
Product
Reactant
Product
Zn(s)|Zn+2(aq)||Cu+2(aq)|Cu(s)
Anode
Salt Bridge
Cathode

ELECTRODE POTENTIAL (E)

- It is the tendency of an electrode to gain/lose e^-.
- It is an interface phenomenon.
- It is measured when a metal is immersed in its own solution.
- It is generally expressed as Reduction Potential.
- Expressed in Volts (V).

STANDARD ELECTRODE POTENTIAL (E°)

- Potential developed at the interface when the metal is immersed in its solution of unit concentration (1M/1N) at 1 atm. pressure and 25°C (298 K).

ELECTROMOTIVE FORCE (EMF)

The difference in potential which causes a current to flow from an electrode of higher potential to that of lower potential.

$$EMF = E_{cell} = E_{cathode} - E_{anode}$$

$$EMF = E^{\circ}_{cell} = E^{\circ}_{cathode} - E^{\circ}_{anode}$$

NERNST EQUATION

An expression of a quantitative relationship between electrode potential and concentration of the electrolyte species in an electro-chemical reaction.

WALTHER HERMANN NERNST
(1864-1941)
NOBEL PRIZE IN CHEMISTRY IN 1920

$$E = E^{0} - \frac{2.303\,RT}{nF} \log Q$$

standard potential

number of electrons

Faraday constant

reaction quotient

POTENTIAL ⟷ **CONCENTRATION**

NERNST EQUATION FOR A CELL

$$M^{n+} + ne^- \rightleftharpoons M$$

$$E = E^o + \frac{0.0591}{n}\log\frac{[reactants]}{[products]}$$

$$E = E^o + \frac{0.0591}{n}\log[M^{n+}]$$

$$Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$$

$$E = E^o + \frac{0.0591}{n}\log\frac{[Cu^{2+}]\cancel{[Zn]}}{[Zn^{2+}]\cancel{[Cu]}}$$

$$E = E^o + \frac{0.0591}{n}\log\frac{[Cu^{2+}]}{[Zn^{2+}]}$$

$$\boxed{E = E^o + \frac{0.0591}{n}\log\frac{[cathode]}{[anode]}}$$

ELECTROCHEMICAL SERIES

*Arrangement of elements in increasing order of **standard electrode potential** values*

- **Less active than H_2**
- **Stronger oxidizing agents**
- **Easy reduction**

HIGHER E°

CATHODE

Elements	Symbol	Electrode Reaction			Electrode Potential
Lithium	Li	$Li^+ (Aq) + e$	—	$Li (s)$	− 3.05
Potassium	K	$K^+ (Aq) + e$	—	$K (s)$	− 2.93
Barium	Ba	$Ba^{2+} (Aq) + 2e$	—	$Ba(s)$	− 2.90
Calcium	Ca	$Ca^{2+} (Aq) + 2e$	—	$Ca(s)$	− 2.87
Sodium	Na	$Na^+ (Aq) + e$	—	$Na (s)$	− 2.71
Magnesium	Mg	$Mg^{2+} (Aq) + 2e$	—	$Mg (s)$	− 2.37
Aluminum	Al	$Al^{3+} (Aq) + 3e$	—	$Al (s)$	− 1.66
Zinc	Zn	$Zn^{2+} (Aq) + 2e$	—	$Zn(s)$	− 0.76
Chromium	Cr	$Cr^{3+} (Aq) + 3e$	—	$Cr (s)$	− 0.74
Iron (2)	Fe	$Fe^{2+} (Aq) + 2e$	—	$Fe (s)$	− 0.44
Cadmium	Cd	$Cd^{2+} (Aq) + 2e$	—	$Cd (s)$	− 0.40
Lead	Pb	$Pb^{2+} (Aq) + 2e$	—	$Pb (s)$	− 0.31
Cobalt	Co	$Co^{2+} (Aq) + 2e$	—	$Co (s)$	− 0.28
Nickel	Ni	$Ni^{2+} (Aq) + 2e$	—	$Ni (s)$	− 0.25
Tin	Sn	$Sn^{2+} (Aq) + 2e$	—	$Sn (s)$	− 0.14
Hydrogen	H_2	$2H^+ + 2e$	—	$H_2 (g)$	0.00
Copper	Cu	$Cu^{2+} (Aq) + 2e$	—	$Cu (s)$	+ 0.34
Iodine	I	$I_2 (s) + 2e$	—	$2 I^- (Aq)$	+ 0.54
Iron (3)	Fe	$Fe^{3+} (Aq) + 3e$	—	$Fe^{2+} (aq)$	+ 0.77
Mercury (1)	Hg	$Hg_2^{2+} (Aq) + 2e$	—	$2 Hg(l)$	+ 0.79
Silver	Ag	$Ag^+ (Aq) + e$	—	$Ag (s)$	+ 0.80
Mercury (2)	Hg	$Hg^{2+} (Aq) + 2e$	—	$Hg (l)$	+ 0.85
Bromine	Br_2	$Br_2 (Aq) + 2e$	—	$2Br^- (Aq)$	+ 1.08
Chlorine	Cl_2	$Cl_2 (Aq) + 2e$	—	$2Cl^- (Aq)$	+ 1.36
Gold	Au	$Au^{3+} (Aq) + 3e$	—	$Au (s)$	+ 1.42
Fluorine	F_2	$F_2 (g) + 2e$	—	$2F^- (aq)$	+ 2.87

ANODE

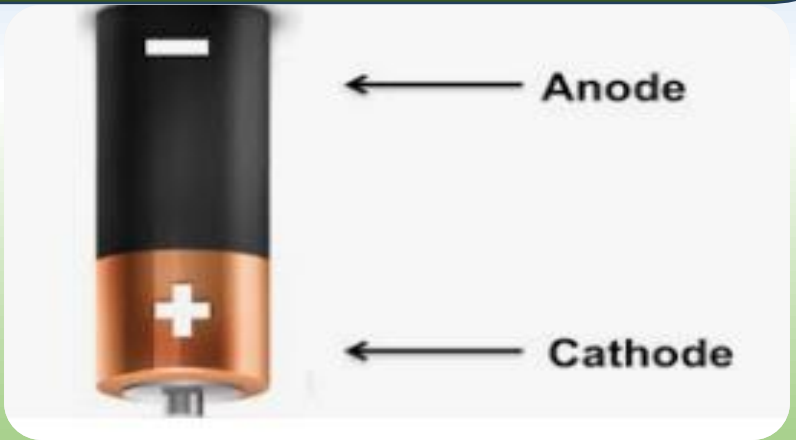
LOWER E°

- **More active than H_2**
- **Stronger reducing agents**
- **Easy oxidation**

ELECTRODE

What is an electrode???

- Good conductor of electricity
- Transfers electrons to non-conducting medium.
- Can be metallic or non-metallic
- Can be reactive or inert
- It is a point where current enters and leaves the system



Types of Electrodes

1. **Metal/Metal ion electrode (M/M^{n+})** :- Metal is in contact with its ionic solution. E.g.:-Cu/Cu^{2+}, Zn/Zn^{2+}, Ag/Ag^{+}

2. **Gas Electrode:-** A gas is in contact with an inert metal dipped in an ionic solution of same gas molecules. E.g.:- Standard Hydrogen electrode

3. **Metal-insoluble metal salt /common ion electrode:-** A metal is in contact with a solution containing the anion of insoluble metal salt. E.g.:- calomel electrode, Ag/AgCl electrode

4. **Ion selective electrode:-** An ion specific membrane is in contact with an ionic solution. e.g., E.g.:- glass electrode, solid state electrode

5. **Redox electrode:-** An inert metal is immersed in a solution containing ions of the same metal having two different oxidation states. E.g: $Pt/Fe^{2+}/Fe^{3+}$

REFERENCE ELECTRODES

REFERENCE ELECTRODE:

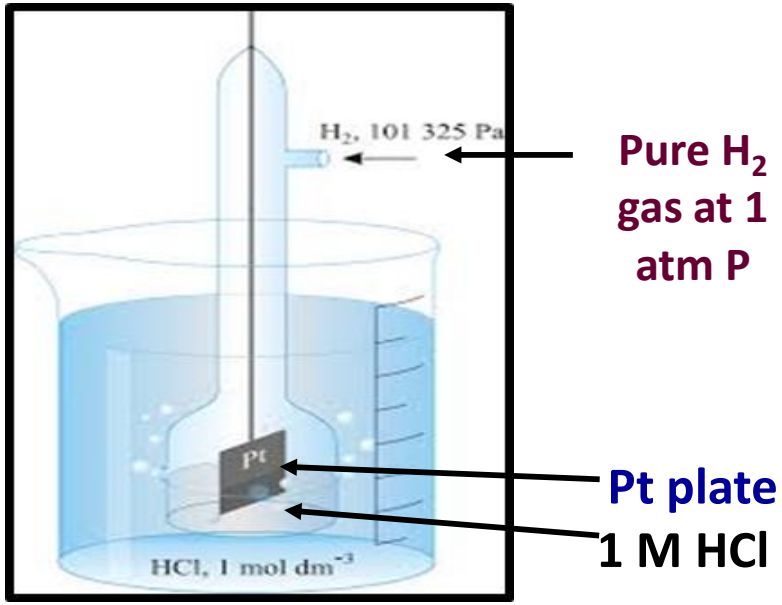
- Constant and reproducible potential
- Used to determine potential of other electrodes

Two types:-

I) Primary Reference Electrode – Standard Hydrogen Electrode (SHE)

Limitations:-

- ✓ Pt is susceptible for poisoning
- ✓ Cannot be used in presence of strong oxidizing and reducing agents.
- ✓ Expensive and difficult to maintain Pt.
- ✓ Portability issue

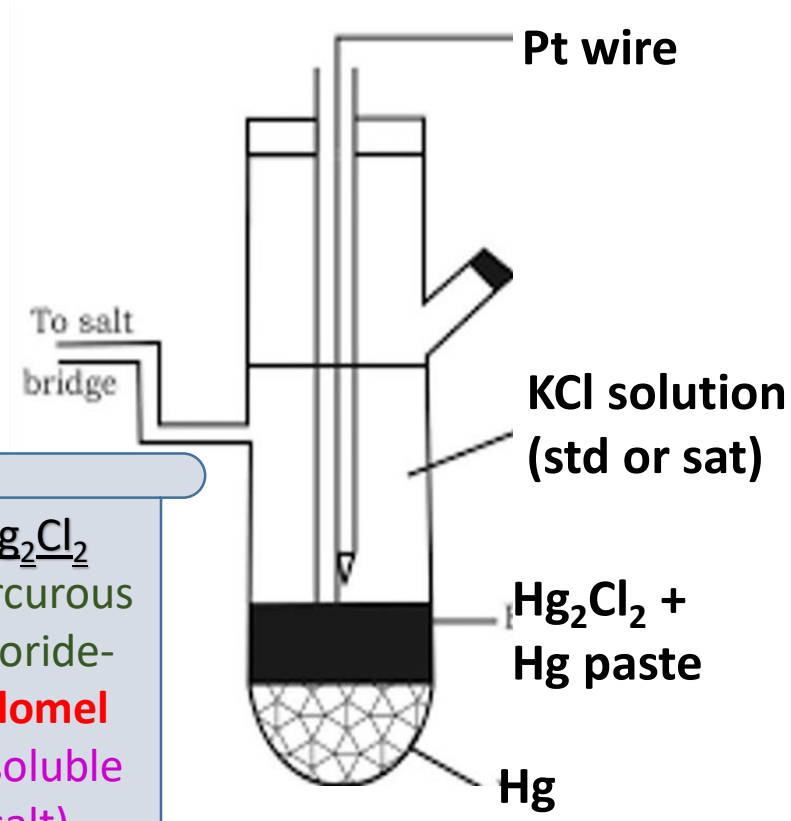


E = 0 Volts

Standard Hydrogen Electrode

CALOMEL ELECTRODE (mercury-mercurous electrode)

Construction



Hg_2Cl_2
Mercurous chloride-
Calomel
(Insoluble salt)

Representation: $Hg_{(l)}|Hg_2Cl_{2(s)}|KCl_{(sat)}$

Working

As anode: $2Hg + 2Cl^- \rightarrow Hg_2Cl_2 + 2e^-$

As cathode: $Hg_2Cl_2 + 2e^- \rightarrow 2Hg + 2Cl^-$

At equilibrium: $Hg_2Cl_2 + 2e^- \rightleftharpoons 2Hg + 2Cl^-$

OR

$^1/_2\, Hg_2Cl_2 + e^- \rightleftharpoons Hg + Cl^-$

Potential: $$E = E^0 - \frac{2.303RT}{F}\log[Cl^-]$$

At 298K, $E_{cal} = E^0_{cal} - 0.0591 \log [Cl^-]$

KCl conc.	0.1 N	1 N	Saturated
E (V)	0.334	0.281	0.2422

Advantages of SCE

Simple Construction

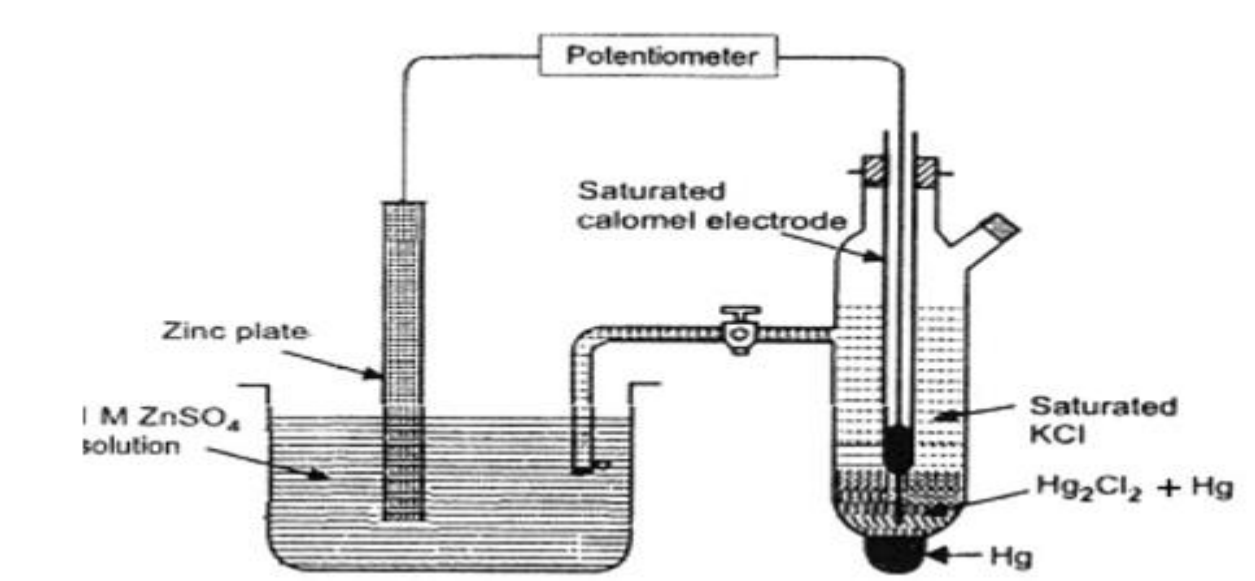
Easily portable

Reproducible potential

Stable potential

Applications of SCE:- i) Determination of Single Electrode Potential
ii)Determination of pH of solution

Determination of Single Electrode Potential using SCE



$$E_{cell} = E_c - E_a$$

$$E_{cell} = E_{SCE} - E_{Zn^{2+}/Zn}$$

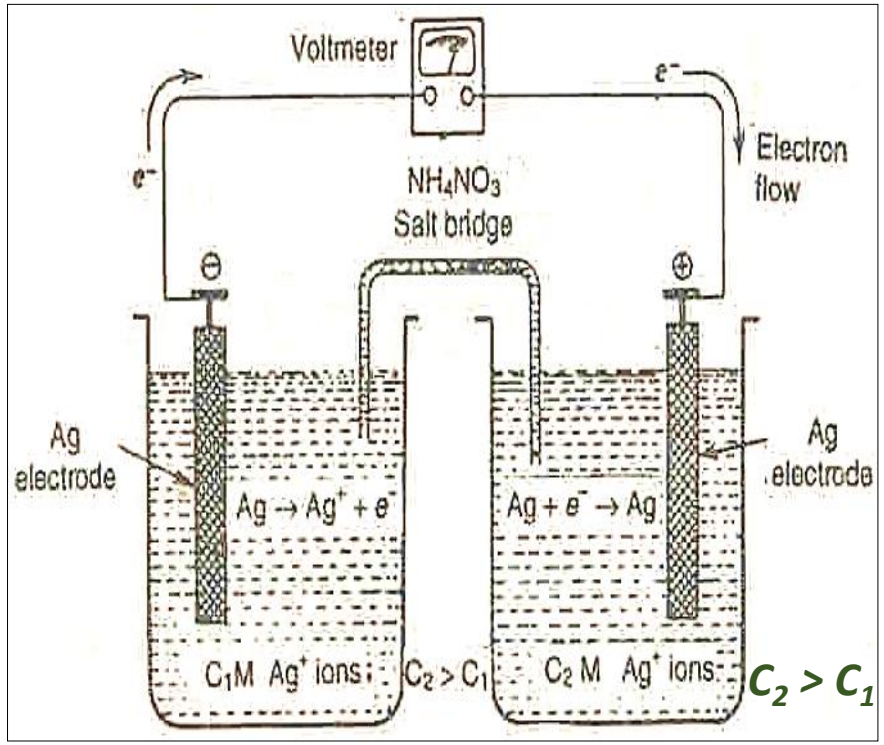
$$E_{cell} = 0.2422 - E_{Zn^{2+}/Zn}$$

$$E_{Zn^{2+}/Zn} = 0.2422 - E_{cell}$$

$Zn_{(s)}|ZnSO_{4(1M)} \parallel KCl_{(sat)}|Hg_2Cl_{2(s)}|Hg_{(l)}$

Concentration cell

An electrochemical cell comprising of two half-cells having identical electrodes and identical electrolytes but with different concentrations, joined by a salt bridge.



Electrode reactions:

Anode : $\cancel{Ag(s)} \rightarrow Ag^+(C_1M) + \cancel{e^-}$

Cathode: $Ag^+(C_2M) + \cancel{e^-} \rightarrow \cancel{Ag(s)}$

Cell reaction : $Ag^+(C_2M) \rightarrow Ag^+(C_1M)$

Equilibrium is created between the two half-cells by transfer of electrons

Emf of the cell (E_{cell}) = $E_{cathode} - E_{anode}$

$$E_{cell} = [E^0 + \frac{RT}{nF}\ln C_2] - [E^0 + \frac{RT}{nF}\ln C_1]$$

$$Ecell = \frac{RT}{nF}\ln[C_2] / \ln[C_1]$$

At 298 K, $$E_{cell} = \frac{0.0591}{n}\log[C_2] / [C_1]$$

Cell Representation:-

$Ag \mid AgNO_3\,(C_1M) \mid\mid AgNO_3\,(C_2M) \mid Ag\ ;\ (C_2 > C_1)$

concentration cell

Low conc. → Anode

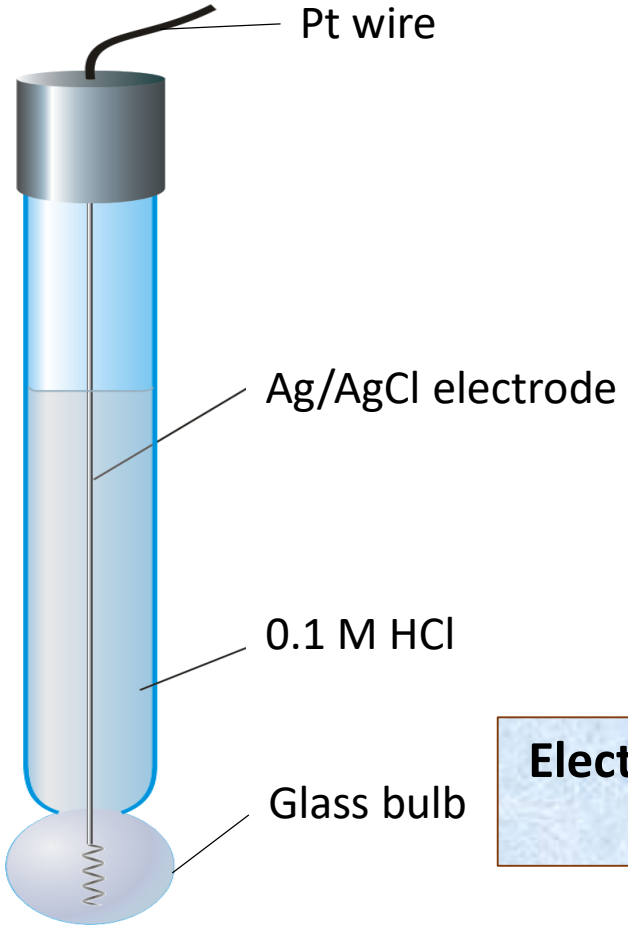
High conc. → Cathode

ION-SELECTIVE ELECTRODE

- They are selective to a particular ion in the solution.
- Their potential depends on the concentration of the selective ion.
- Also referred to as membrane electrode.

GLASS ELECTRODE

- Sensitive to H^+ ions
- Also called as pH electrode

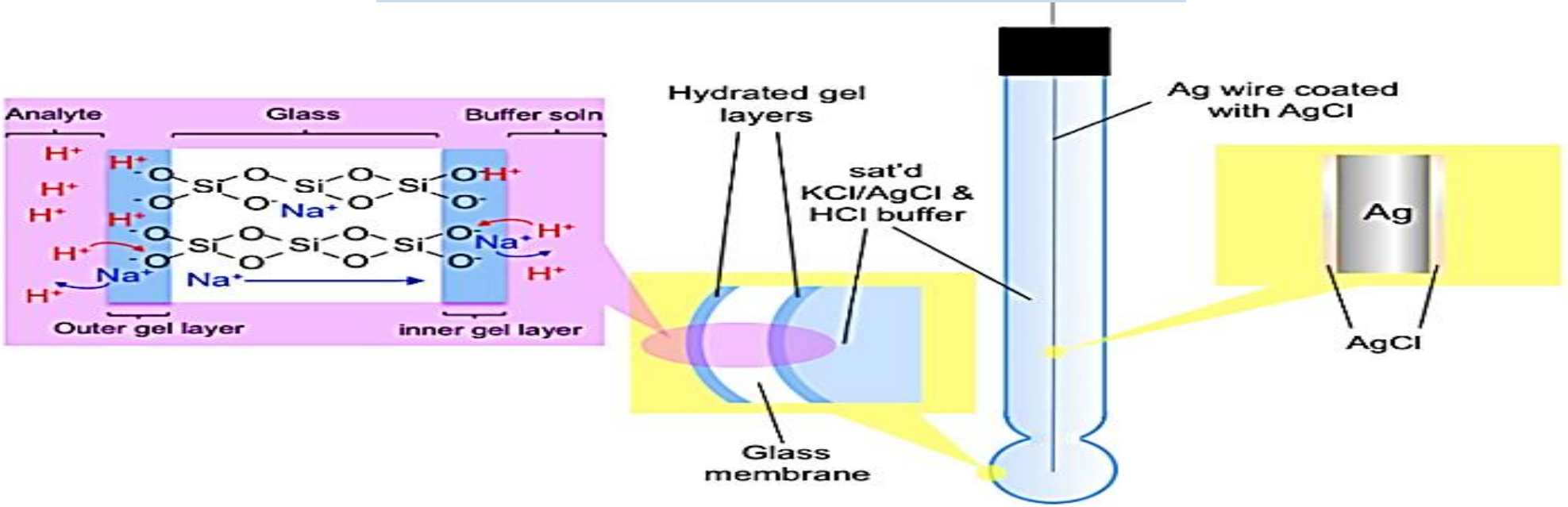


Secondary reference electrode
Internal reference electrode

Representation:
$Ag_{(s)}|AgCl_{(s)}|HCl_{(0.1M)}|glass$

Electrically conductive glass – Corning Glass
6% CaO; 72% SiO_2; 22% Na_2O

GLASS ELECTRODE



Working:

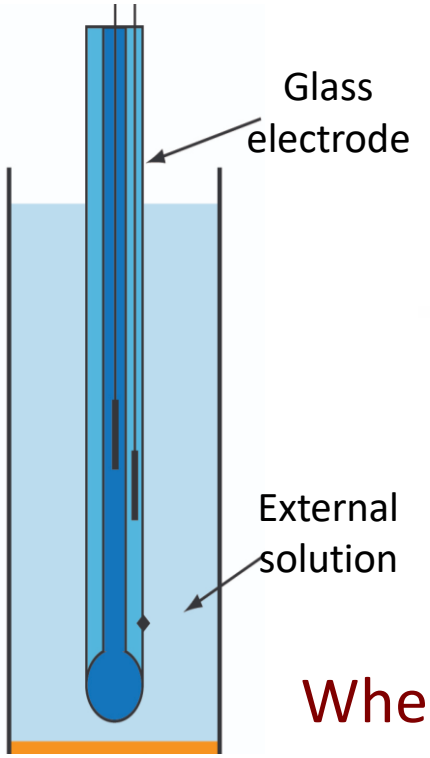
- The glass is a partially hydrated aluminosilicate containing Na^+/Ca^{2+} ions.
- If a thin-walled glass bulb containing an acid is immersed in another solution, the membrane undergoes an **ion exchange reaction**.
- The Na^+ ions on the glass membrane are exchanged for H^+ ions in the solution at the hydrated layer.

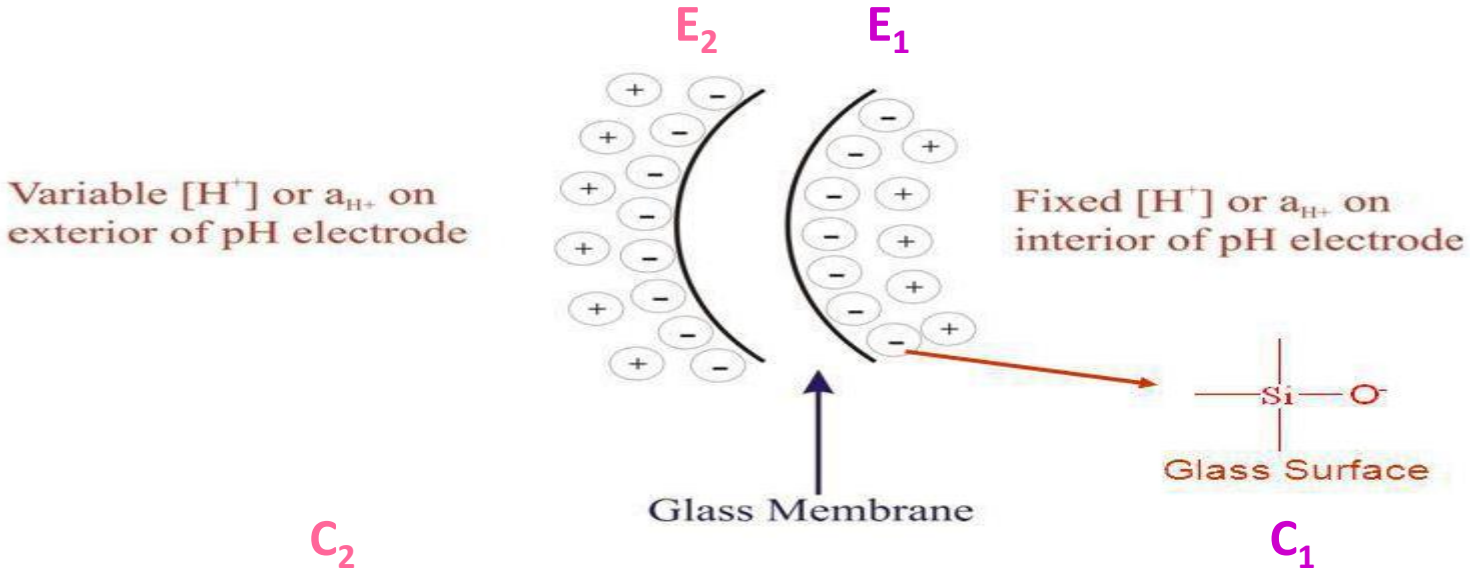
$$H^+ + Na^+Gl^- \leftrightharpoons Na^+ + H^+Gl^-$$

Working of a Glass Electrode

The overall potential of the glass electrode depends upon three parameters

1) The boundary potential (E_b)
2) The potential of the internal Ag/AgCl reference electrode. ($E_{Ag/AgCl}$)
3) A small unpredictable contribution called the asymmetry potential, (E_{asym})





When immersed in an external acidic solution, at the membrane-solvent interface, a small local potential develops called **BOUNDARY POTENTIAL.**

$$E_b = E_2 - E_1 = 0.0591 \log C_2 - 0.0591 \log C_1$$

$$E_b = 0.0591 \log C_2 - 0.0591 \log C_1$$

C_1 is a constant

$$E_b = K + 0.0591 \log C_2$$

$$K = -0.0591 \log C_1$$

$$E_b = K + 0.0591 \, log \, [H^+]$$

$$pH = -\log[H^+]$$

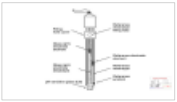
$$E_b = K - 0.0591 \, pH$$

$$E_g = E_b + E_{Ag/AgCl} + E_{asym}$$

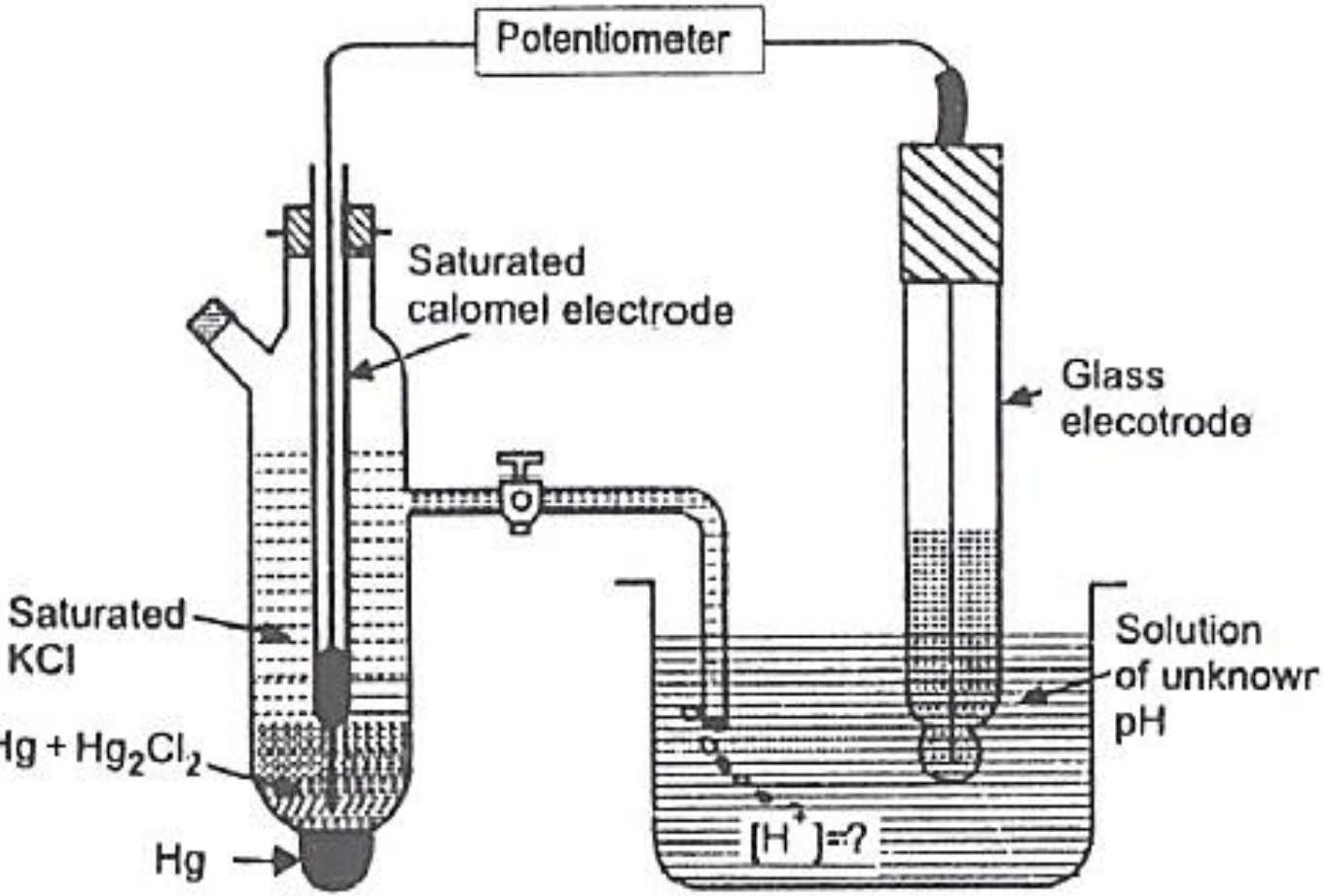
$$E_g = K - 0.0591pH + E_{Ag/AgCl} + E_{asym}$$

$$\boxed{E_g = E_g^0 - 0.0591pH}$$

$$E_g^0 = K + E_{Ag/AgCl} + E_{asym}$$



Determination of pH using glass electrode



$$E_{cell} = E_c - E_a$$

$$E_{cell} = E_g - E_{cal}$$

$$E_{cell} = [E_g^0 - 0.0591pH] - E_{cal}$$

$$pH = \frac{E_g^0 - E_{cal} - E_{cell}}{0.0591}$$

$Hg_{(l)}|Hg_2Cl_{2(s)}|KCl_{(sat)}\ ||H^+_{(x\ M)}|glass|HCl_{(0.1M)}|AgCl_{(s)}|Ag_{(s)}$

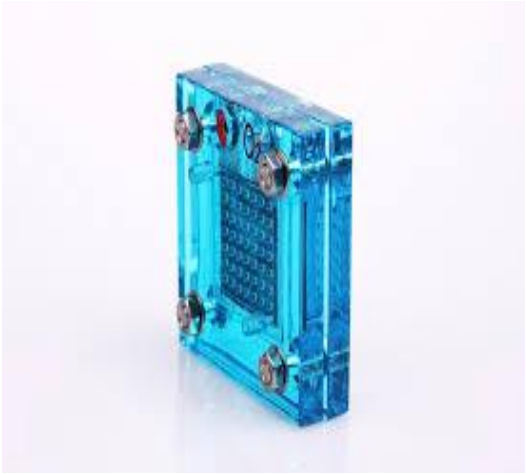
GLASS ELECTRODE

ADVANTAGES

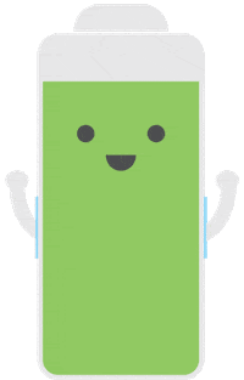
- Employed in the presence of strong oxidizing and reducing substances and metal ions.
- Simple and portable.
- Does not get poisoned easily.
- Effective in the pH range 1-9.However, by using special glass electrodes, pH 1-13 can be measured.

DISADVANTAGES

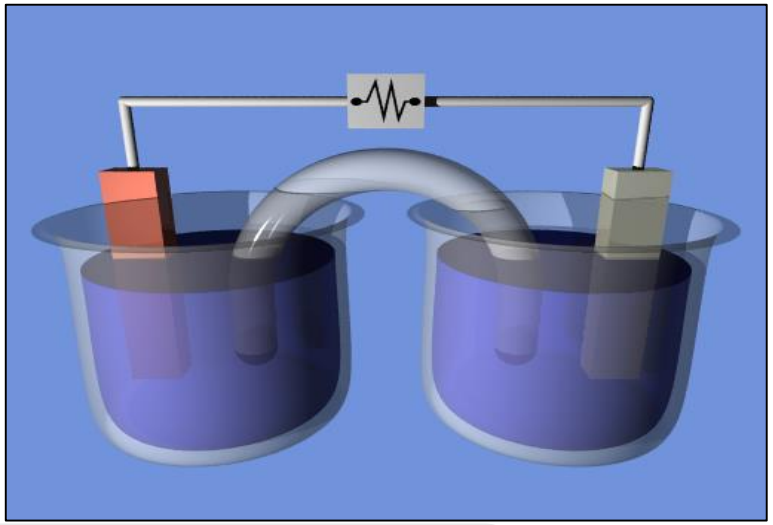
- Ineffective in highly acidic (pH<1) and highly alkaline (pH>11) solutions.
- Not responsive in presence of pure alcohols.

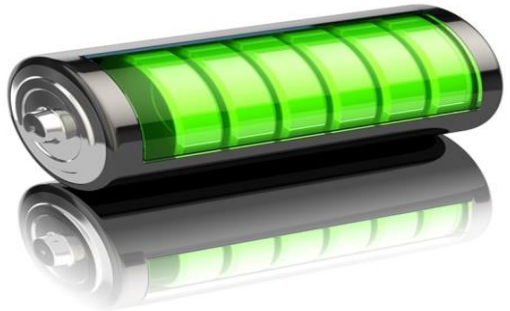


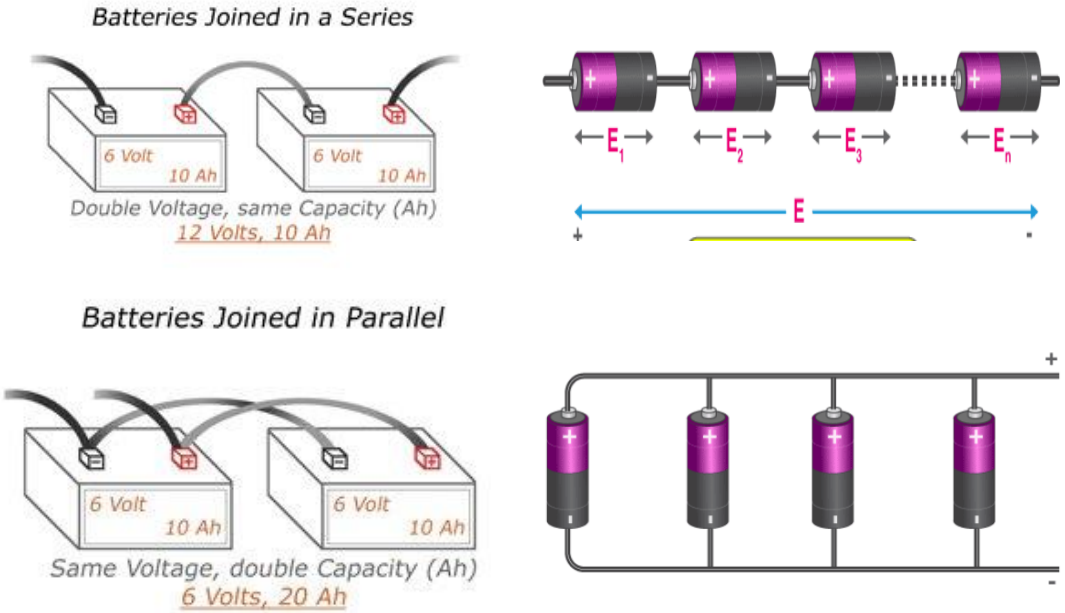
ENERGY SYSTEMS

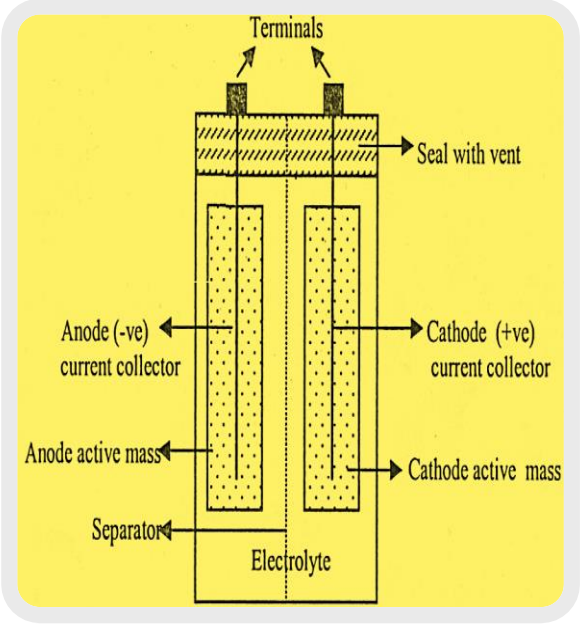


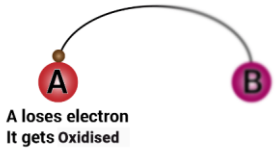
BATTERIES





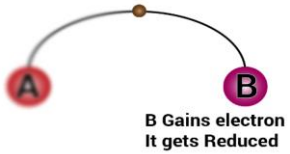






Anode

Cathode



Electrolyte

Separator

Battery components selection criteria

Anode Material

- Ease of oxidation
- High coulombic output
- Good conductivity
- High stability
- Ease of fabrication

Electrolyte

- Good ionic conductor but bad electrical conductor to prevent internal short circuit
- Resistance to the electrode material
- Safety for handling

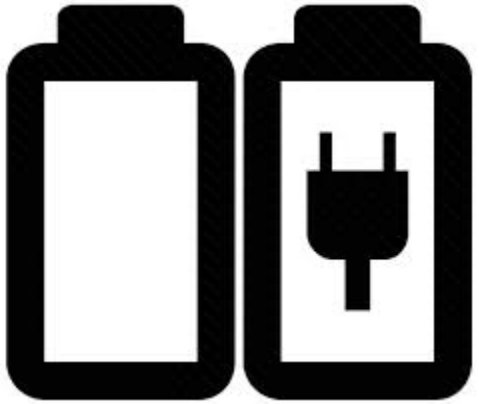
Separator

- Permeable
- Capacity to prevent mixing
- Good ionic conductor but bad electrical conductor
- High strength flexibility
- Resistant to the active materials

Cathode Material

- Ease of reduction of active species
- Resistance to electrolyte

TYPES OF BATTERIES



Dry Cell
Zinc air battery

Lead acid battery
Lithium ion battery

PRIMARY BATTERY

Non-rechargeable batteries
Cell reactions *cannot* be reversed

SECONDARY BATTERY

Rechargeable batteries
Cell reactions *can* be reversed

$$A + B \rightarrow C + D$$

$$A + B \rightleftarrows C + D$$

TYPES OF BATTERIES

RESERVE BATTERY

Deactivated until use/ A component is isolated.

Prior to usage, the component is added/activated to generate electricity for immediate use.

A kind of Primary Battery.

Also called Stand-by Battery.

APPLICATIONS AND EXAMPLES

Used in military, aerospace applications

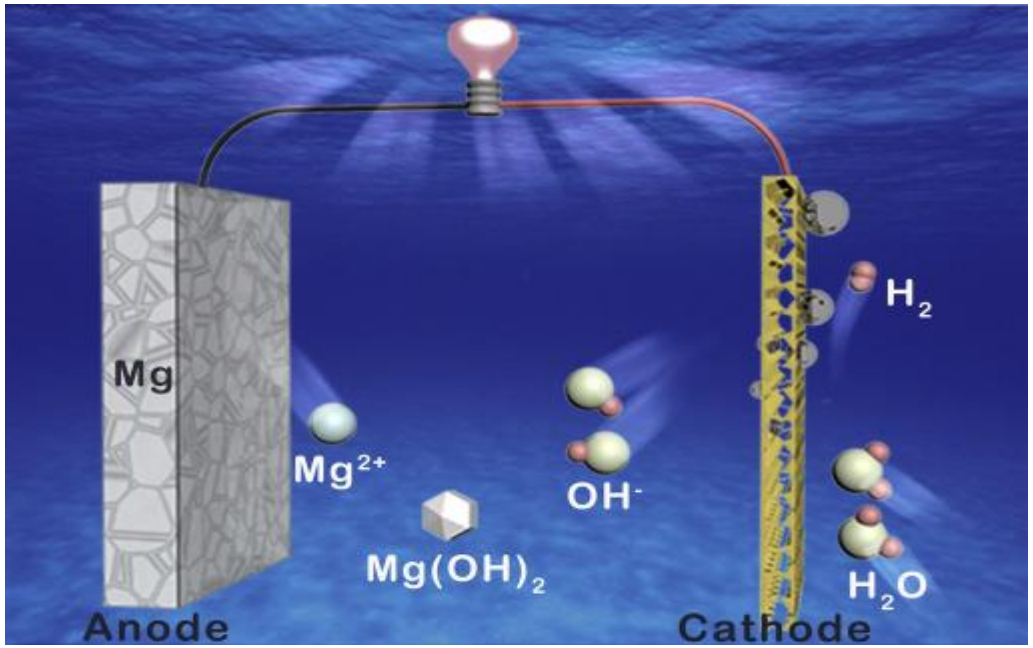
Example:

1) Water activated Mg battery

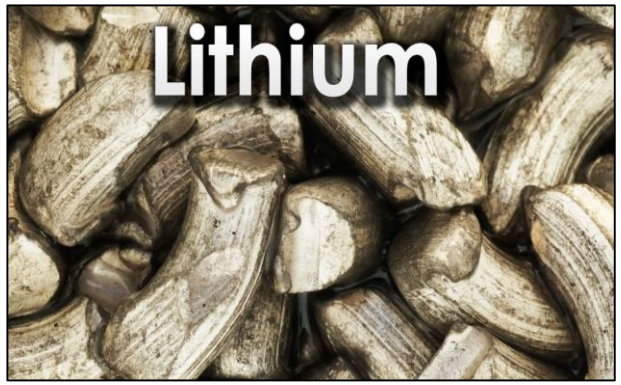
Activated by adding water (electrolyte)

2) Thermal battery (Li/FeS_2)

Activated by heating the molten electrolyte



LITHIUM-ION BATTERY



Light weight

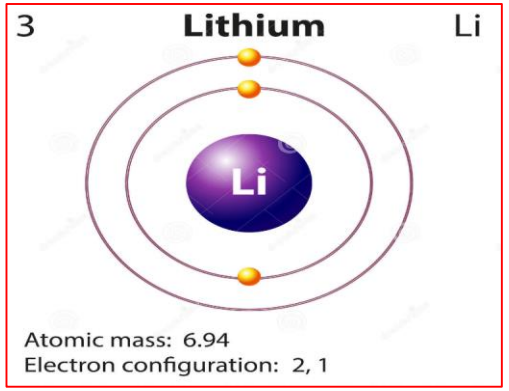
WHY LITHIUM???

Low electrode potential (-3V)

Increase in energy density and power density.

High reactivity

High conductivity



Quicker oxidation and faster transport of electrons.

LITHIUM-ION BATTERY

ANODE:

Li intercalated in graphite layers

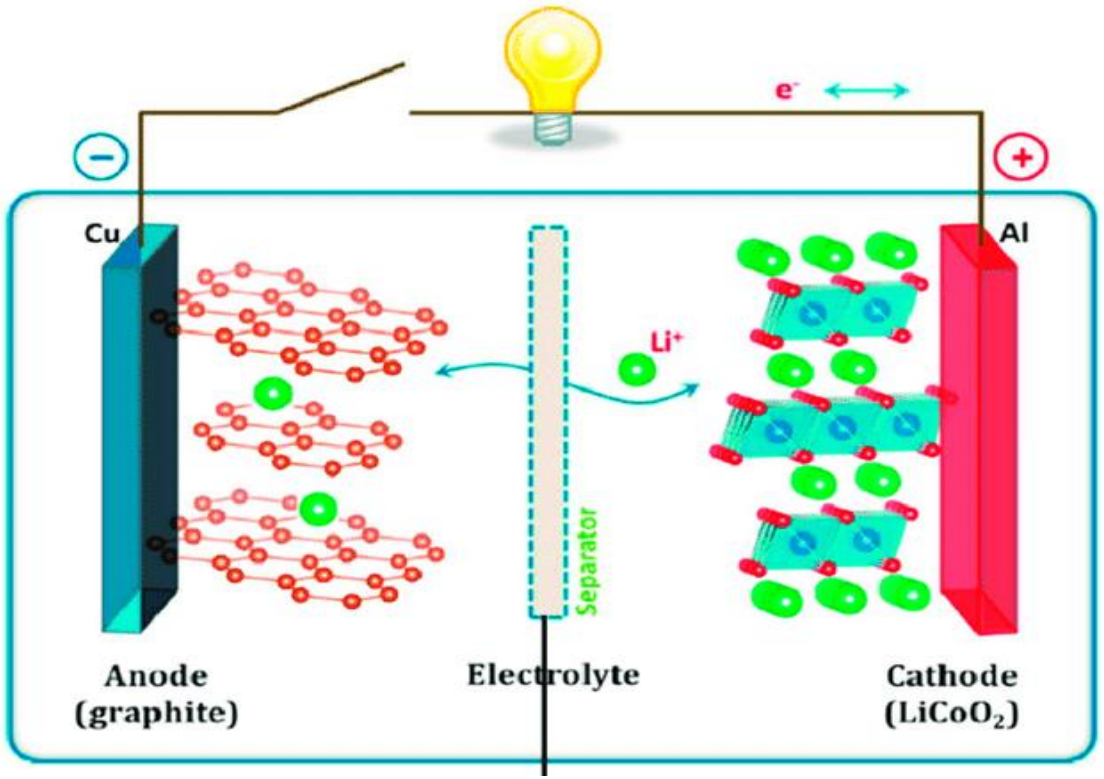
ELECTROLYTE:

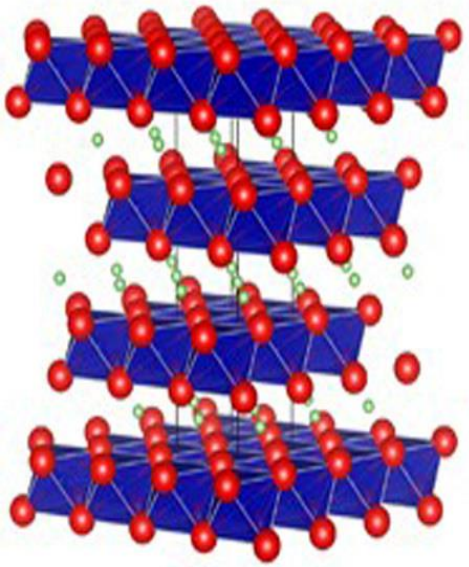
Li salt in organic solvent

($LiPF_6$ in ethylene carbonate)

SEPARATOR:

Porous polyethylene film





CATHODE:

$Li_{1-x}CoO_2$

(partially lithiated cobalt oxide)

Anode $$Li_xC \longrightarrow xLi^+ + C + xe^-$$

Cathode $$Li_{1-x}CoO_2 + xLi^+ + xe^- \longrightarrow LiCoO_2$$

Net reaction $$Li_xC + Li_{1-x}CoO_2 \longrightarrow LiCoO_2 + C$$

3.6 V

LITHIUM-ION BATTERY

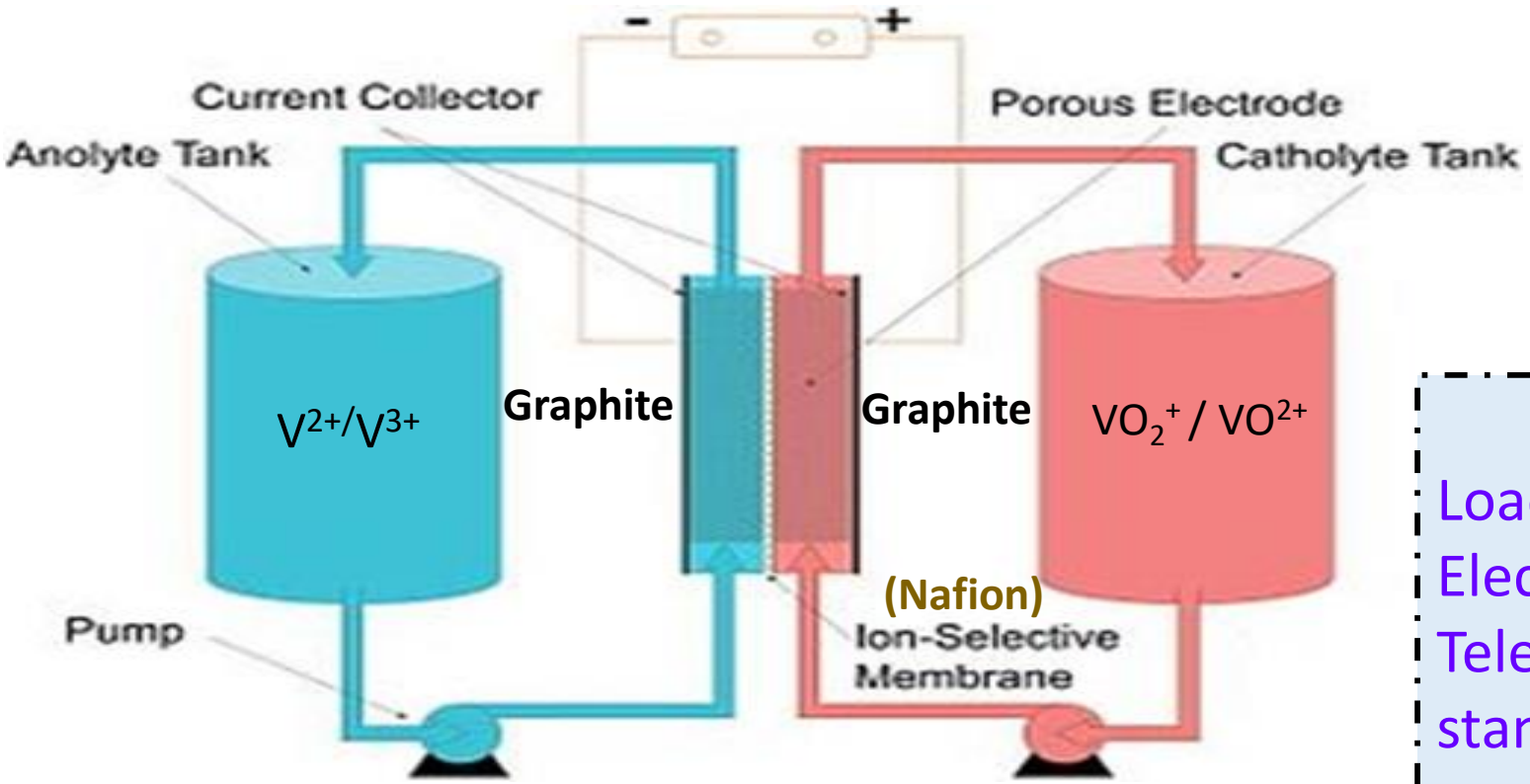
Applications:

- Mobile phones
- Laptops
- Telecommunication equipment's
- Aerospace applications
- Electric vehicles

FLOW BATTERY

- Energy is stored in the form of electrolyte.
- Electrolyte can be stored in electrolyte tanks.
- Electrolyte needs to be continuously supplied.
- Flow batteries are “rechargeable” as the tank can be re-filled with fresh electrolyte.
- High efficiency batteries.

VANADIUM REDOX FLOW BATTERY (VRFB)



APPLICATIONS

Load Balancing, Electric vehicles, UPS, Telecommunication, stand alone power systems, Military equipment's

Anode:

$$V^{2+} \rightarrow V^{3+} + e^-$$

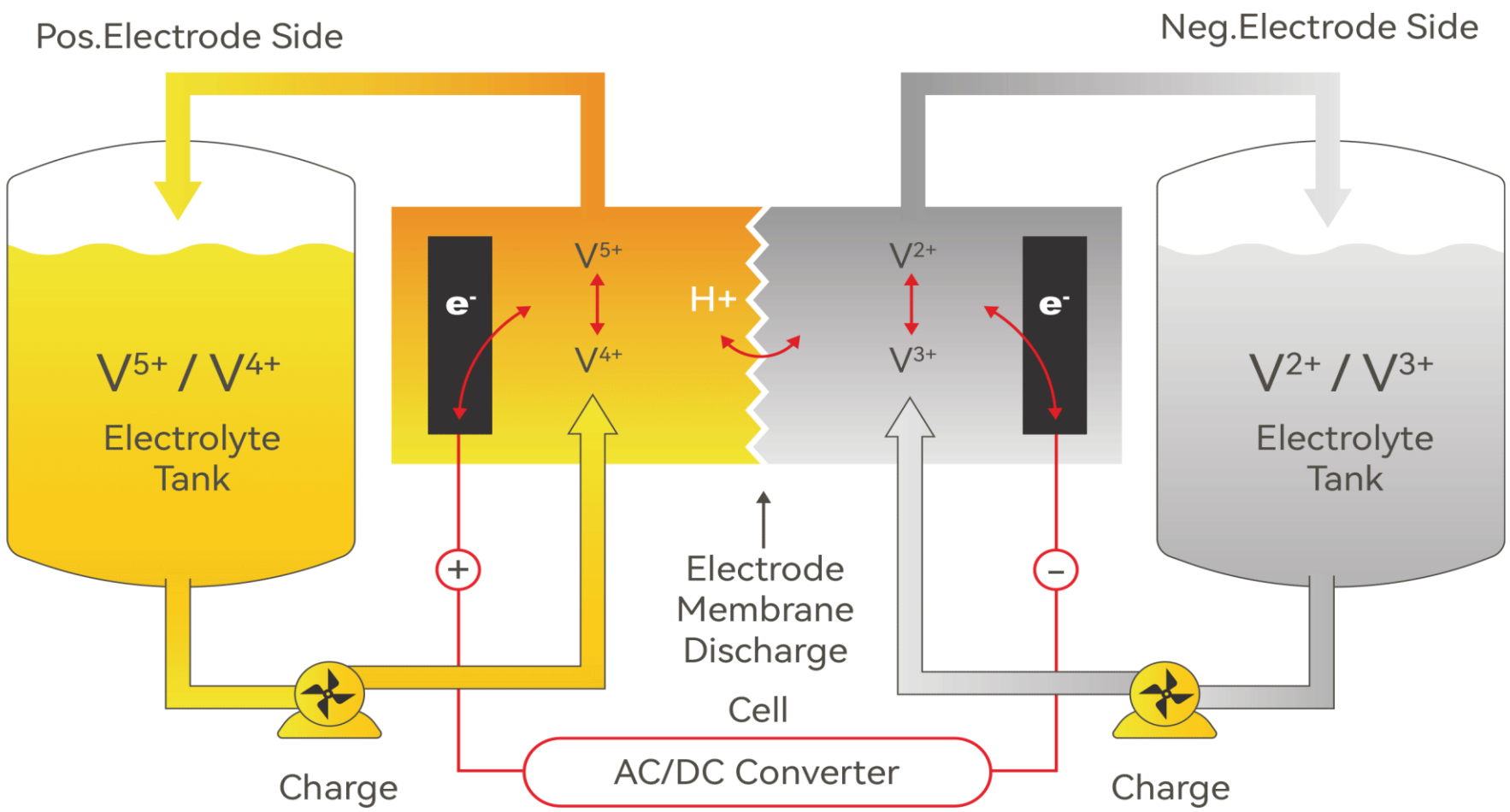
Cathode:

$$VO_2^+ + 2H^+ + e^- \rightarrow VO^{2+} + H_2O$$

NCR:

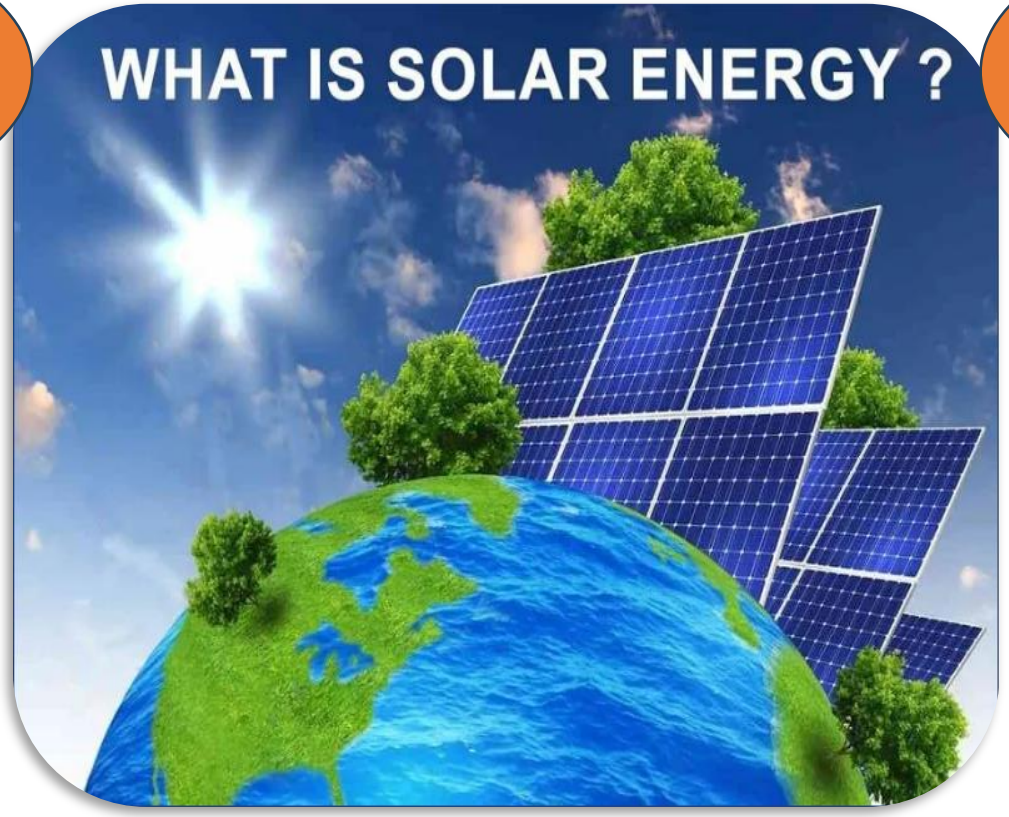
$$V^{2+} + VO_2^+ + 2H^+ \rightarrow V^{3+} + VO^{2+} + H_2O$$

1.26 V


Pos.Electrode Side
Neg.Electrode Side
V^{5+} / V^{4+}
Electrolyte Tank
e^-
V^{5+}
V^{4+}
H+
V^{2+}
V^{3+}
e^-
V^{2+} / V^{3+}
Electrolyte Tank
+
−
Electrode Membrane Discharge Cell
Charge
AC/DC Converter
Charge

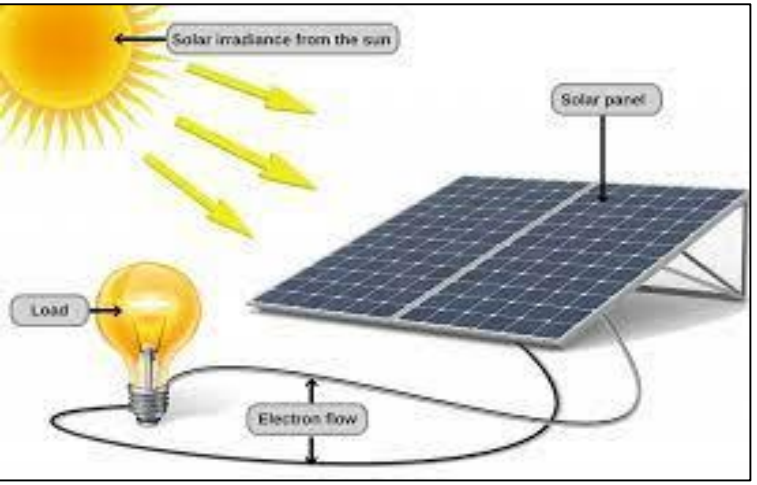
SOLAR ENERGY

Most abundant

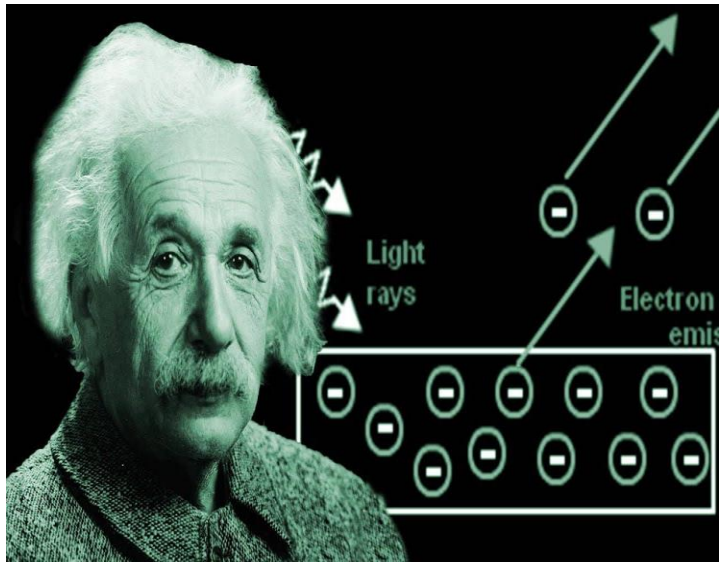


Free & Renewable

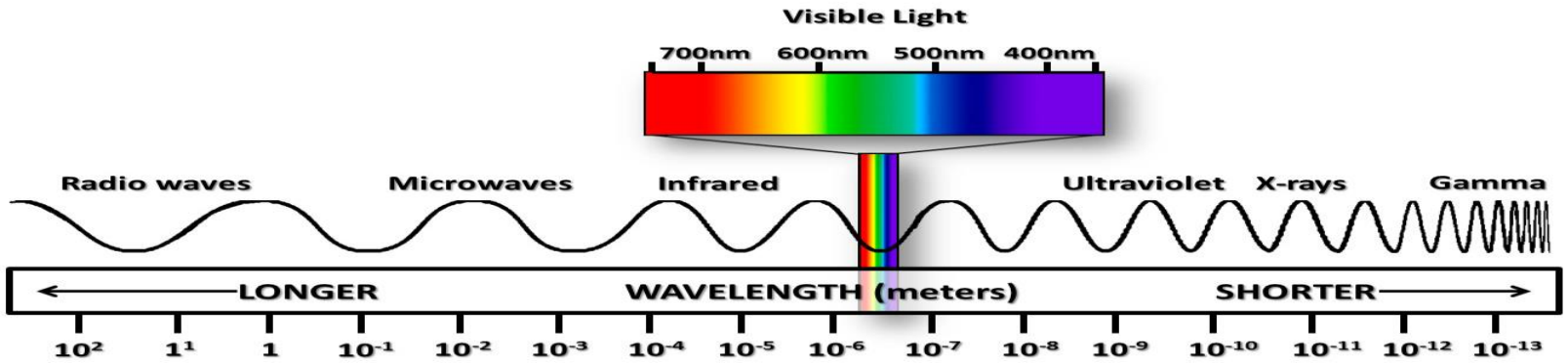
It is the radiating light and heat from the sun that is harnessed using devices like heaters, solar cookers, and photovoltaic cells to convert it to electrical energy and heat.



SOLAR CELL/PHOTOVOLTAIC CELL (PV CELL)



- Building blocks of solar panel.
- Made of semiconductors which have the capacity to absorb light.
- Converts solar energy to electrical energy.
- Works on the principle of photovoltaic effect ($E = hc/\lambda$).



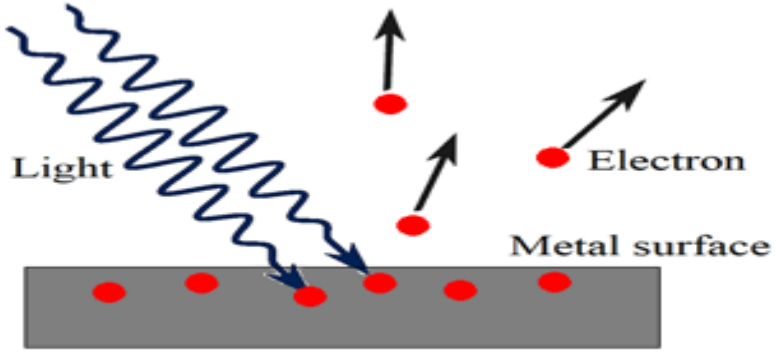
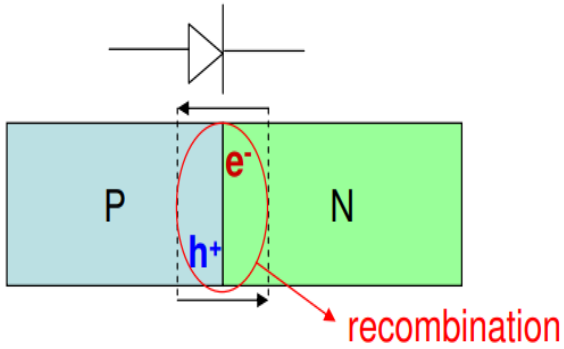


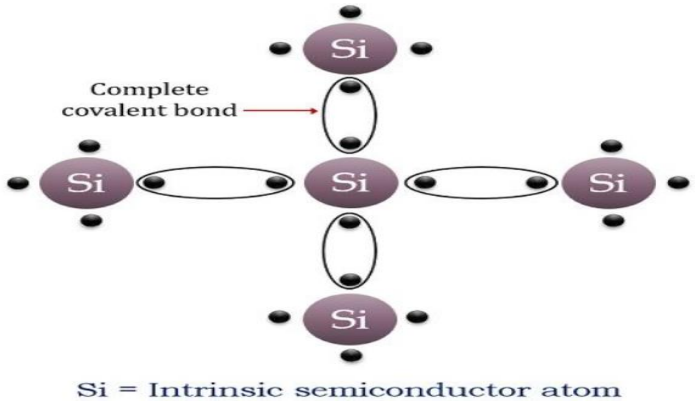
Fig. 1. Photoelectric effect

PHOTOVOLTAIC CELL

Silicon is an intrinsic semiconductor.
It has 4 electrons in its outer shell.
Designed with p-n junction :- p-n junction diode

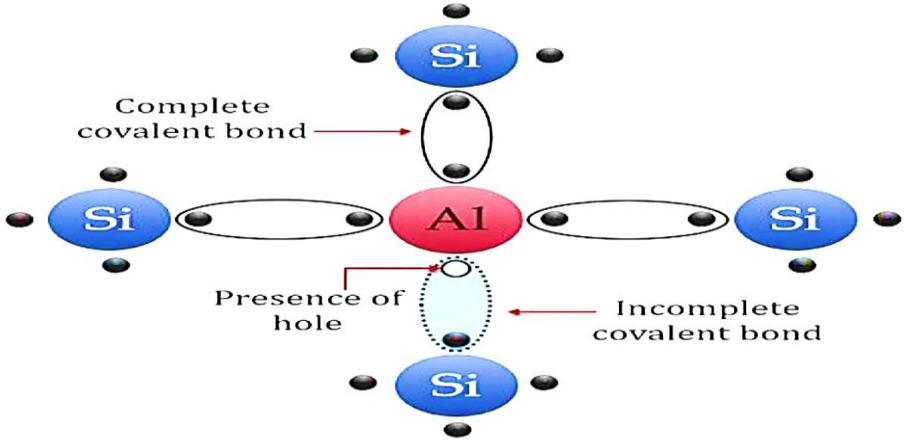


p-n diode



p-type semiconductor

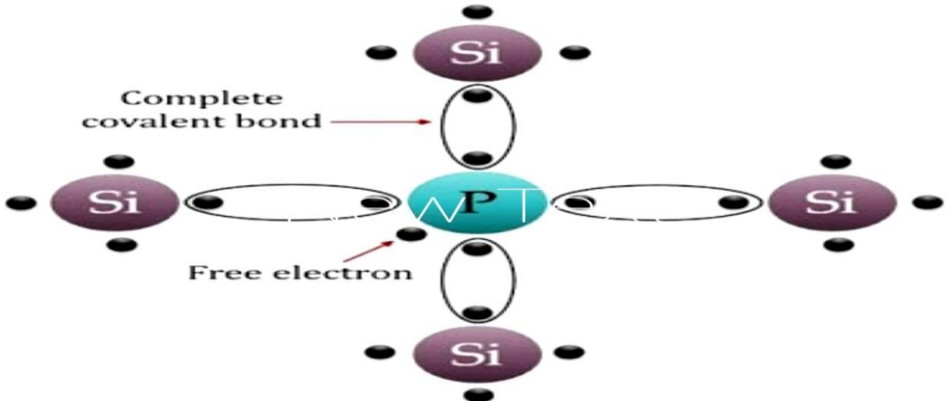
- Formed by doping Si with atoms having 3 electrons in outer shell (eg. B, Ga, Al).
- Has excess of positive holes.



p-type semiconductor

n-type semiconductor

- Formed by doping Si with atoms having 5 electrons in outer shell (eg. P, Sb).
- Has excess of negative electrons.

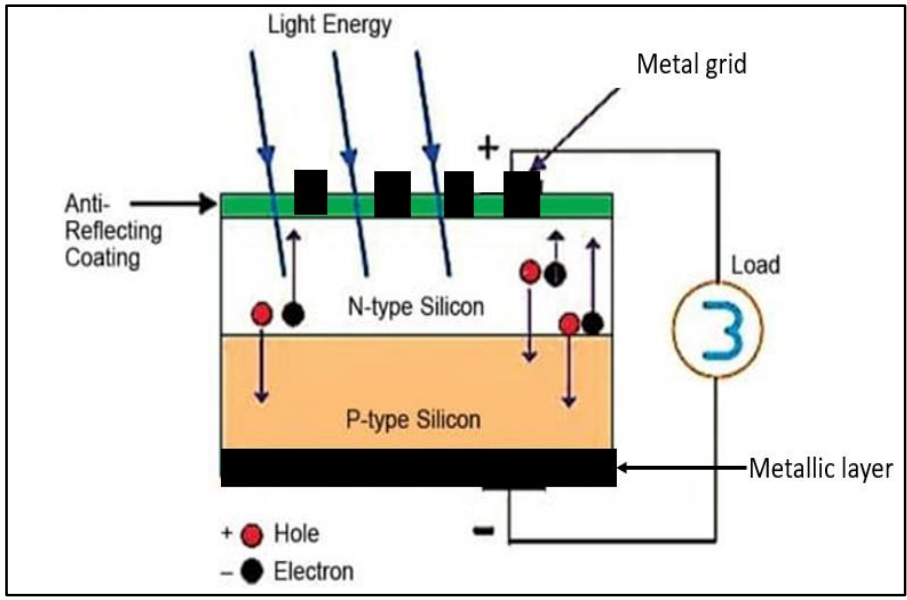


n-type semiconductor

Construction and Working of solar Photovoltaic cell

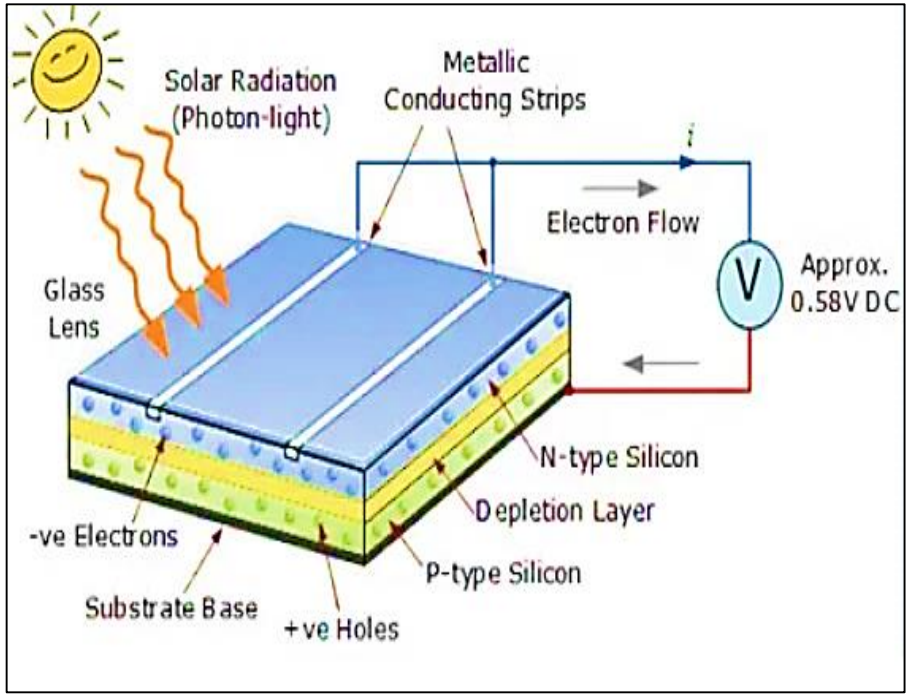
Construction:-

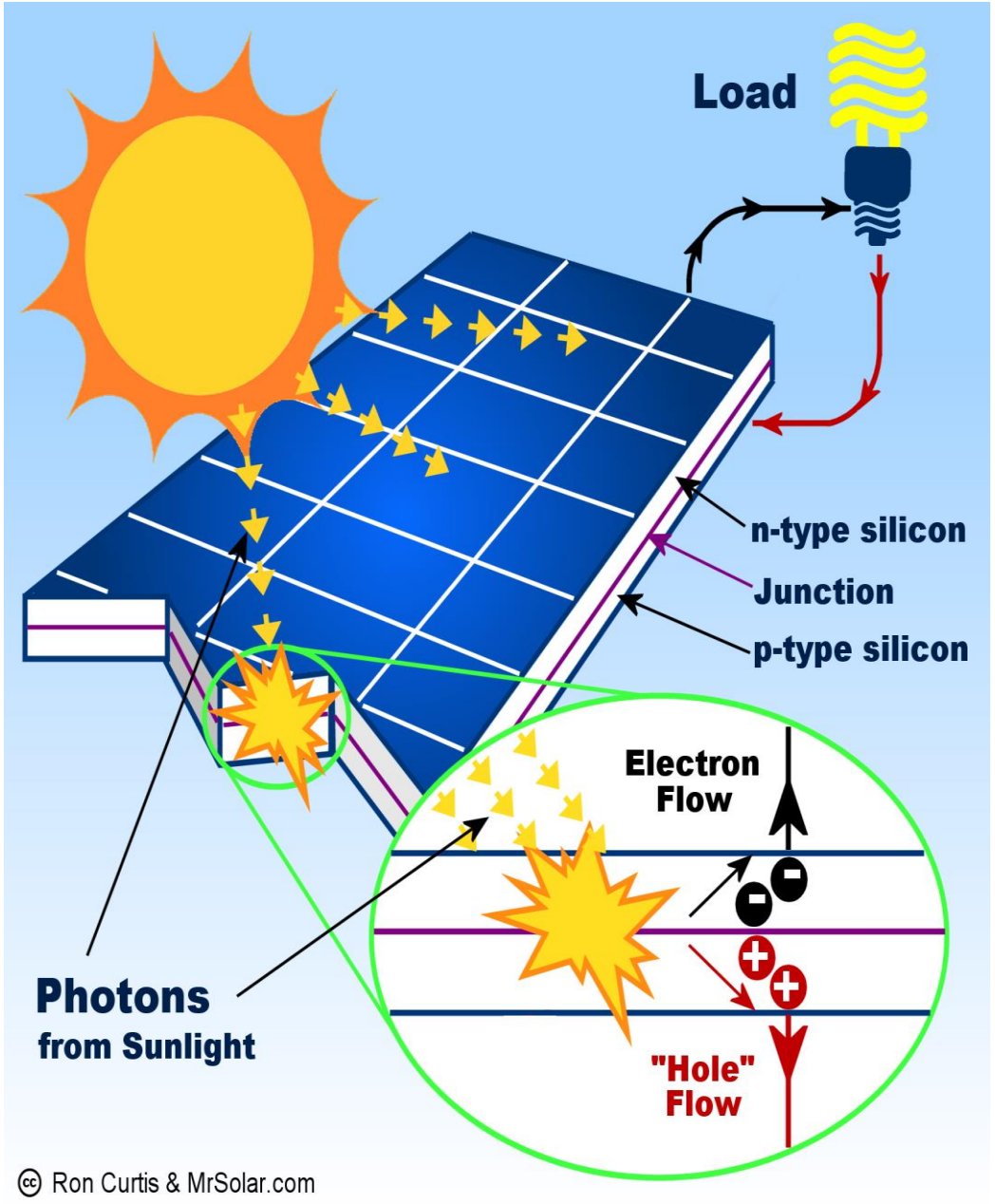
- A PV cell is composed of a thin poly crystalline silicon wafer consisting of an ultra-thin layer of n-type Si on top of p-type Si.
- Thus, a p-n junction is formed.
- A proper metal contacts are made on the n-type side and p-type side of the S.C for electrical contact.
- An antireflective layer between the grid lines increases the amount of light transmitted to the semiconductor.



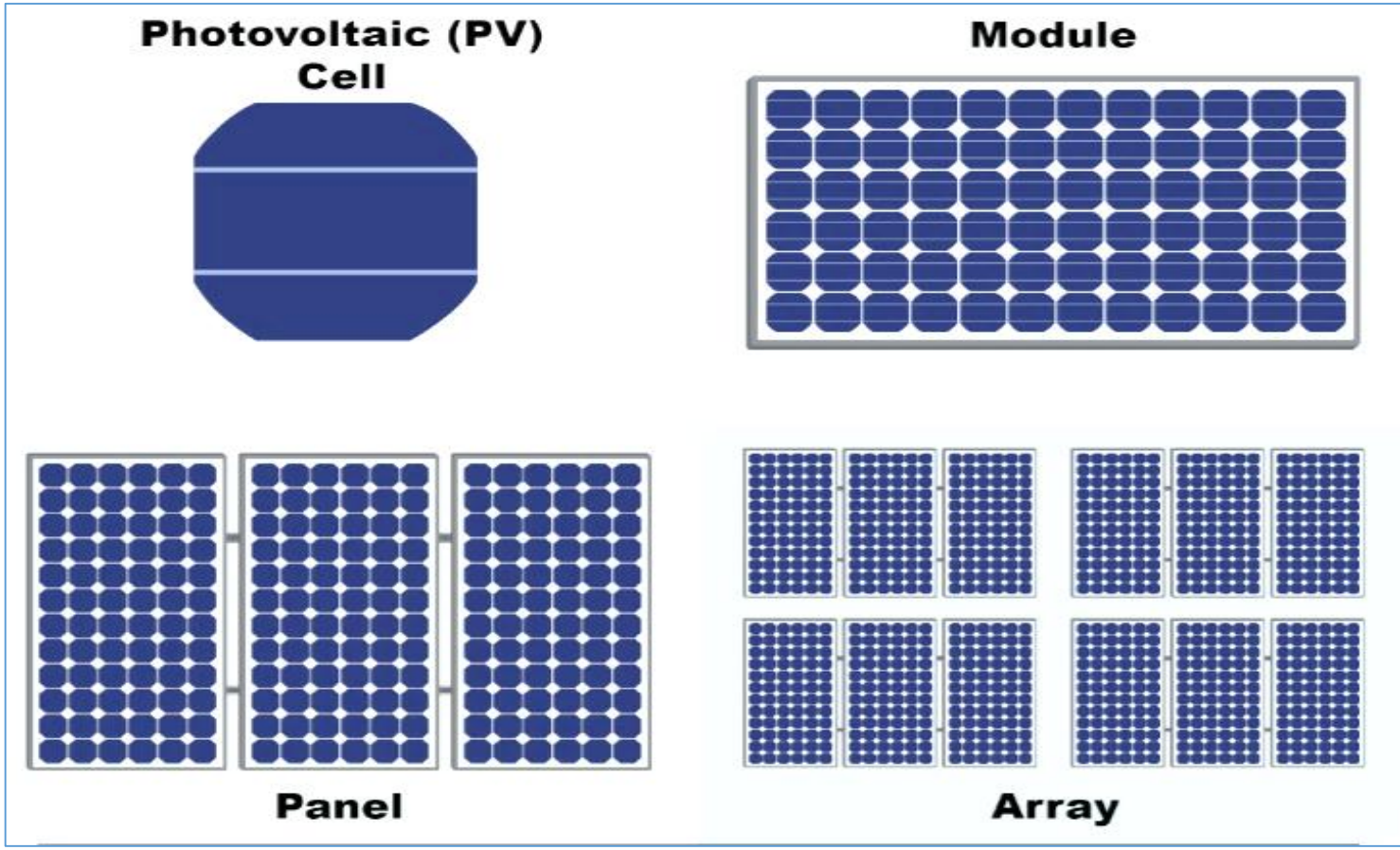
Working

- PV cell works on the principle of photoelectric effect, E = hγ.
- When light radiation falls on the p-n junction diode ,if the incident energy is greater than the band-gap of the S.C, electrons get excited and are released to form electron-hole pairs.
- The electrons are drifted to and collected at the n-type end and the holes are drifted to p-type end.
- Photoelectric current is produced by the flow of current between the two ends through the external circuit.





The power generated by a single photovoltaic cell is typically only about 0.5-1 V. But by connecting large numbers of individual cells or solar panel arrays, thousands of kilowatts of electric power can be generated.



Applications of PV's

- Electricity for parking meters,
- Temporary traffic signs,
- Emergency phones,
- Radio transmitters,
- Water irrigation pumps,
- Stream-flow gauges,
- Remote guard posts,
- Lighting for roadways

Satellite Panels

ADVANTAGES	DISADVANTAGES
• Solar power is pollution free • Low operating costs (no fuels) • Facilities can operate with little maintenance • Solar power is economical except the initial set up. • High public acceptance and excellent safety record.	• Sunlight is a diffuse source. i.e. It is relatively low-density energy. • High installation costs. • Energy can be produced only during the day time. • Space required to generate power output is relatively more. • Generation of DC current, hence requires a converter (DC to AC).

Production of Solar Grade Silicon (Union Carbide Process)

Step 1: SiO_2 is reduced with carbon (coke or charcoal) at 1500-2000℃ to produce metallurgical grade silicon (98% purity).

$$SiO_2 + C \xrightarrow{1500 - 2000^oC} Si + 2CO$$

Step 2: The metallurgical grade silicon is treated with HCl at 300℃ to form trichlorosilane.

$$Si + 3HCl \xrightarrow{300^oC} HSiCl_3 + H_2$$

Production of Solar Grade Silicon (Union Carbide Process)

Step 3: Trichlorosilane is then passed through fixed bed columns containing ammonium ion exchange resin catalyst, where trichlorosilane converts to silicon hydride.

$$2HSiCl_3 \rightarrow H_2SiCl_2 + SiCl_4$$

$$3H_2SiCl_2 \rightarrow SiH_4 + 2HSiCl_3$$

Step 4: Silicon hydride gas is purified by distillation and pyrolyzed at 600 - 900℃ to form high purity semiconductor grade polysilicon.

$$SiH_4 \xrightarrow{600\ -900^oC} Si + 2H_2$$

THANK YOU